6.2: Eukaryotic cells have internal membranes that compartmentalize their functions

Cells—the basic structural and functional units of every organism—are of two distinct types: prokaryotic and eukaryotic. Organisms of the domains Bacteria and Archaea consist of prokaryotic cells. Organisms of the domain Eukarya—protists, fungi, animals, and plants—all consist of eukaryotic cells. ("Protist" is an informal term referring to a diverse group of mostly unicellular eukaryotes.)

Comparing Prokaryotic and Eukaryotic Cells

All cells share certain basic features: They are all bounded by a selective barrier, called the *plasma membrane* (or the cell membrane). Inside all cells is a semifluid, jellylike substance called **cytosol**, in which subcellular components are suspended. All cells contain *chromosomes*, which carry genes in the form of DNA. And all cells have *ribosomes*, tiny complexes that make proteins according to instructions from the genes.

A major difference between prokaryotic and eukaryotic cells is the location of their DNA. In a **eukaryotic cell**, most of the DNA is in an organelle called the *nucleus*, which is bounded by a double membrane (see Figure 6.8). In a **prokaryotic cell**, the DNA is concentrated in a region that is not membrane-enclosed, called the **nucleoid(Figure 6.5)**. *Eukaryotic* means "true nucleus" (from the Greek *eu*, true, and *karyon*, kernel, referring to the nucleus), and *prokaryotic* means "before nucleus" (from the Greek *pro*, before), reflecting the earlier evolution of prokaryotic cells.

Figure 6.5

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A prokaryotic cell.
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A prokaryotic cell.

Lacking a true nucleus and the other membrane-enclosed organelles of the eukaryotic cell, the prokaryotic cell appears much simpler in internal structure. Prokaryotes include bacteria and archaea; the general cell structure of these two domains is quite similar.

Figure a is a diagram of a prokaryotic cell, a typical rod-shaped bacterium, and it is compared to figure b, a TEM of a thin section through the bacterium Bacillus coagulans (scale 0.5 micrometers). Structures are labeled on the bacterial diagram and TEM as follows:

- Fimbriae: (shown on the diagram only) hair-like attachment structures on the surface of some prokaryotes, here cover the surface of the cell.
- Nucleoid: region where the cell's DNA is located (not enclosed by a membrane). In the diagram, a large region composed of a tangled string-like structure that is the bacterial chromosome. In the TEM, a dark region in the middle of the cell is visible.
- Ribosomes: complexes that synthesize proteins. In both the diagram and the TEM seen as small dots.
- Plasma membrane: a membrane enclosing the cytoplasm, just inside the cell wall.
- Cell wall: a rigid structure outside the plasma membrane.
- Capsule: jellylike outer coating of many prokaryotes; the outer portion of both the diagram and the TEM.
- Flagella: long, whip-like structures extending from one end of the bacterial cell in the diagram. Not shown in the TEM. Locomotion structures of some bacteria.

Animation

Prokaryotic Cell Structure and Function

image

image

The interior of either type of cell is called the **cytoplasm**; in eukaryotic cells, this term refers only to the region between the nucleus and the plasma membrane. Within the cytoplasm of a eukaryotic cell, suspended in cytosol, are a variety of organelles of specialized form and function. These membrane-bounded structures are absent in almost all prokaryotic cells, another distinction between prokaryotic and eukaryotic cells. In spite of the absence of organelles, though, the prokaryotic cytoplasm is not a formless soup. For example, some prokaryotes contain regions surrounded by proteins (not membranes), within which specific reactions take place.

Eukaryotic cells are generally much larger than prokaryotic cells (see Figure 6.2). Size is a general feature of cell structure that relates to function. The logistics of carrying out cellular metabolism sets limits on cell size. At the lower limit, the smallest cells known are bacteria called mycoplasmas, which have diameters between 0.1 and 1.0 μm. These are perhaps the smallest packages with enough DNA to program metabolism and enough enzymes and other cellular equipment to carry out the activities necessary for a cell to sustain itself and reproduce. Typical bacteria are 1–5 μm in diameter, about ten times the size of mycoplasmas. Eukaryotic cells are typically 10–100 μm in diameter.

Metabolic requirements also impose theoretical upper limits on the size that is practical for a single cell. At the boundary of every cell, the **plasma membrane** functions as a selective barrier that allows passage of enough oxygen, nutrients, and wastes to service the entire cell **(Figure 6.6)**. For each square micrometer of membrane, only a limited amount of a particular substance can cross per second, so the ratio of surface area to volume is critical. As a cell (or any other object) increases in size, its surface area grows proportionately less than its volume. (Area is proportional to a linear dimension squared, whereas volume is proportional to the linear dimension cubed.) Thus, a smaller cell has a greater ratio of surface area to volume: Compare the calculations for the first two "cells" in **Figure 6.7**. The **Scientific Skills Exercise** gives you a chance to calculate the volumes and surface areas of two actual cells—a mature yeast cell and a cell budding from it. To explore different ways that the surface area of cells is maximized in various organisms, see Make Connections Figure 33.8.

Figure 6.6

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The plasma membrane.
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The plasma membrane.

The plasma membrane and the membranes of organelles consist of a double layer (bilayer) of phospholipids with various proteins attached to or embedded in it. The hydrophobic parts of phospholipids and membrane proteins are found in the interior of the membrane, while the hydrophilic parts are in contact with aqueous solutions on either side. Carbohydrate side chains may be attached to proteins or lipids on the outer surface of the plasma membrane.

Figure a. TEM of a plasma membrane. The plasma membrane, here in a red blood cell, appears as a pair of dark bands separated by a light band. (Scale: 0.1 micrometers).

Figure b. Structure of the plasma membrane, consisting of hydrophilic and hydrophobic regions. Hydrophilic regions exist on both sides of the membrane, consisting of a mat of spherical polar heads of phospholipids. The hydrophobic region is the center of the bilayer, with the lipid tails of the phospholipids appearing as parallel lines tucked into the interior of the membrane. Membrane proteins are embedded in the membrane spanning both layers, while carbohydrate side chains stick out from the surface of the cell.

VISUAL SKILLS *What parts of the membrane diagram in (b) correspond to the dark bands in the TEM in (a)? What parts correspond to the gold band? (Review*Figure 5.11*.)**

*For suggested answer, see*Appendix A*.**

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Membranes

image

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Figure 6.7

Geometric relationships between surface area and volume.

Geometric relationships between surface area and volume.

A cube with sides of 5 units has a surface area that increases while the total volume remains constant as it is divided into smaller cubes with sides of 1 unit. Each face of the larger cube is 5 by 5.

A table is as follows:

Total surface area [sum of the surface areas (height times width) of all box sides times number of boxes]	**6**	**150**
Total volume [height times width times length times number of boxes]	1	125
Surface-to-volume (S-to-V) ratio [surface area divided by volume]	6	12

In this diagram, cells are represented as cubes. Using arbitrary units of length, we can calculate the cell's surface area (in square units, or units2), volume (in cubic units, or units3), and ratio of surface area to volume. A high surface area-to-volume ratio facilitates the exchange of materials between a cell and its environment.

Figure 6.7 Walkthrough

Geometric relationships between surface area and volume

Watch an explanation of this figure in the Figure 6.7 Walkthrough: Geometric relationships between surface area and volume.

image

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Scientific Skills Exercise

Using a Scale Bar to Calculate Volume and Surface Area of a Cell

AP Science Practices 2, 5

How Much New Cytoplasm and Plasma Membrane Are Made by a Growing Yeast Cell? The unicellular yeast *Saccharomyces cerevisiae* divides by budding off a small new cell that then grows to full size (see the yeast cells at the bottom of Figure 6.8). During its growth, the new cell synthesizes new cytoplasm, which increases its volume, and new plasma membrane, which increases its surface area. In this exercise, you will use a scale bar to determine the sizes of a mature parent yeast cell and a cell budding from it. You will then calculate the volume and surface area of each cell. You will use your calculations to determine how much cytoplasm and plasma membrane the new cell needs to synthesize to grow to full size.

How the Experiment Was Done Yeast cells were grown under conditions that promoted division by budding. The cells were then viewed with a differential interference contrast light microscope and photographed.

Data from the Experiment This light micrograph shows a budding yeast cell about to be released from the mature parent cell:

A cell buds from a mature parent cell; the budding cell is smaller and appears fused with the parent cell on one side. Scale is 1 micrometer.

Micrograph from K. Tatchell, using yeast cells grown for experiments described in L. Kozubowski et al., Role of the septin ring in the asymmetric localization of proteins at the mother-bud neck in *Saccharomyces cerevisiae, Molecular Biology of the Cell* 16:3455–3466 (2005).

Micrograph from K. Tatchell, using yeast cells grown for experiments described in L. Kozubowski et al. Role of the septin ring in the asymmetric localization of proteins at the mother-bud neck in Saccharomyces cerevisiae, Molecule Biology of the Cell 16:3455-3466 (2005).

INTERPRET THE DATA

- Examine the micrograph of the yeast cells. The scale bar under the photo is labeled 1μm. The scale bar works in the same way as a scale on a map, where, for example, 1 inch equals 1 mile. In this case the bar represents one thousandth of a millimeter. Using the scale bar as a basic unit, determine the diameter of the mature parent cell and the new cell. Start by measuring the scale bar and the diameter of each cell. The units you use are irrelevant, but working in millimeters is convenient. Divide each diameter by the length of the scale bar and then multiply by the scale bar's length value to give you the diameter in micrometers.
- The shape of a yeast cell can be approximated by a sphere.

- - Calculate the volume of each cell using the formula for the volume of a sphere:

V=43πr3

A sphere's diameter is a segment from one boundary to another, passing through the center. The radius is the distance from one boundary to the center.

- - Note that π (the Greek letter pi) is a constant with an approximate value of 3.14, d stands for diameter, and r stands for radius, which is half the diameter.

- - What volume of new cytoplasm will the new cell have to synthesize as it matures? To determine this, calculate the difference between the volume of the full-sized cell and the volume of the new cell.

- As the new cell grows, its plasma membrane needs to expand to contain the increased volume of the cell.
- Calculate the surface area of each cell using the formula for the surface area of a sphere: A=4πr2.
- How much area of new plasma membrane will the new cell have to synthesize as it matures?
- When the new cell matures, it will be approximately how many times greater in volume and how many times greater in surface area than its current size?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

The need for a surface area large enough to accommodate the volume helps explain the microscopic size of most cells and the narrow, elongated shapes of some cells, such as nerve cells. Larger organisms do not generally have *larger* cells than smaller organisms—they simply have *more* cells (see the far right of Figure 6.7). A sufficiently high ratio of surface area to volume is especially important in cells that exchange a lot of material with their surroundings, such as intestinal cells. Such cells may have many long, thin projections from their surface called *microvilli*, which increase surface area without an appreciable increase in volume.

The evolutionary relationships between prokaryotic and eukaryotic cells will be discussed later in this chapter, and prokaryotic cells will be described in detail elsewhere (see Chapter 27). Most of the discussion of cell structure that follows in this chapter applies to eukaryotic cells.

A Panoramic View of the Eukaryotic Cell

In addition to the plasma membrane at its outer surface, a eukaryotic cell has extensive, elaborately arranged internal membranes that divide the cell into compartments—the organelles mentioned earlier. The cell's compartments provide different local environments that support specific metabolic functions, so incompatible processes can occur simultaneously in a single cell. The plasma membrane and organelle membranes also participate directly in the cell's metabolism because many enzymes are built right into the membranes.

The basic fabric of most biological membranes is a double layer of phospholipids and other lipids. Embedded in this lipid bilayer or attached to its surfaces are diverse proteins (see Figure 6.6). However, each type of membrane has a unique composition of lipids and proteins suited to that membrane's specific functions. For example, enzymes embedded in the membranes of the organelles called mitochondria function in cellular respiration. Because membranes are so fundamental to the organization of the cell, Chapter 7 will discuss them in detail.

Before continuing with this chapter, examine the eukaryotic cells in **Figure 6.8**. The generalized diagrams of an animal cell and a plant cell introduce the various organelles and show the key differences between animal and plant cells. The micrographs at the bottom of the figure give you a glimpse of cells from different types of eukaryotic organisms.

Figure 6.8

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Exploring Eukaryotic Cells
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A cutaway of a generalized animal cell.

Parts of the cell are labeled as follows.

- Nucleus. Consists of the nuclear envelope enclosing the nucleolus and chromatin. The nucleus comprises nearly 1/3 of the cell in this diagram.
- Nuclear envelope: double membrane enclosing the nucleus; perforated by pores; continuous with E R.
- Nucleolus: nonmembranous structure involved in production of ribosomes; a nucleus has one or more nucleoli.
- Chromatin: material consisting of D N A and proteins; visible in a dividing cell as individual condensed chromosomes.
- Endoplasmic reticulum (E R): network of membranous sacs and tubes; active in membrane synthesis and other synthetic and metabolic processes; has rough (ribosome-studded) and smooth regions. (Rough E R) Continuous with the nuclear envelope, the rough E R appears as many folded

layers studded with dot-like ribosomes. (Smooth E R) Beyond the rough E R, the smooth E R is much smaller (about 1/5 the size of the rough E R) and has a thick web-like appearance in this diagram.

- Golgi apparatus: organelle active in synthesis, modification, sorting, and secretion of cell products. (Golgi apparatus) Between the rough E R and the plasma membrane, the Golgi apparatus is also drawn as a heavily folded structure similar in size to the smooth E R.
- Mitochondrion: organelle where cellular respiration occurs and most A T P is generated. (Mitochondrion) Kidney bean shaped structures, about half the size of the Golgi apparatus, drawn in cross section so that the inner folds (cristae) are visible. Several shown.
- Lysosome: digestive organelle where macromolecules are hydrolyzed. Small spherical structure about one-fourth the size of the mitochondria.
- Peroxisome: organelle with various specialized metabolic functions; produces hydrogen peroxide as a by-product and then converts it to water. Small spherical structure similar in size to the lysosome.
- Ribosomes (small brown dots): complexes that make proteins; free in cytosol or bound to rough E R or nuclear envelope.
- Cytoskeleton: reinforces cell's shape; functions in cell movement; components are made of protein. Includes microfilaments, intermediate filaments, and microtubules. (Cytoskeleton) Drawn as long tubules or thick filaments or string-like structures and composed in order of decreasing thickness: the microtubules, the intermediate filaments, and the microfilaments.
- Centrosome: region where the cell's microtubules are initiated; contains a pair of centrioles.
- Plasma membrane: membrane enclosing the cell.
- Flagellum. Appears as a whip-like structure protruding from the cell; length is similar to cell length as drawn.
- Flagellum: motility structure present in some animal cells, composed of a cluster of microtubules within an extension of the plasma membrane.
- Microvilli: projections that increase the cell's surface area.
- Animal cells. Human cells from the lining of the uterus (colorized TEM) with nucleus about 10 micrometers in diameter and nucleolus visible as a dark spot within the nucleus.

A cutaway view of a generalized plant cell.

Parts of the plant cell are labeled as follows.

- Chloroplast: photosynthetic organelle; converts energy of sunlight to chemical energy stored in sugar molecules. Composed of the microfilaments and the thicker microtubules.

- Plasmodesmata: cytoplasmic channels through cell walls that connect the cytoplasms of adjacent cell.
- Cell wall: outer layer that maintains cell's shape and protects cell from mechanical damage; made of cellulose, other polysaccharides, and protein.
- Central vacuole. Drawn here in the center of the cell, a large structure comparable in size to the nucleus. Central vacuole: prominent organelle in older plant cells; functions include storage, breakdown of waste products, and hydrolysis of macromolecules; enlargement of the vacuole is a major mechanism of plant growth.
- Golgi apparatus. Drawn here between the central vacuole and the plasma membrane, the Golgi apparatus is a heavily folded structure about one-third the size of the central vacuole.
- Nucleus. The nuclear envelope encloses the nucleolus and chromatin.
- Plant cells. Cells from duckweed (Spirodela oligorrhiza), a floating plant (colorized TEM). Cell walls separate cells containing distinct nuclei with nucleoli, chloroplasts and mitochondria. Scale: 5 micrometers.

Two micrographs are as follows.

- Unicellular Eukaryotes. Colorized SEM: Unicellular green alga Chlamydomonas (cell diameter = about 8 micrometers) with two flagella visible.
- Unicellular Eukaryotes. Colorized TEM (scale: 1 micrometer): In slice view labeled are the nucleus with dark nucleolus, six chloroplasts, a vacuole, cell wall, and two flagella.

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Tour of an Animal Cell

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Tour of a Plant Cell

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Video

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Chlamydomonas

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